SINGLE-PHASE · LOW-VOLTAGE BATTERY · 3 – 6 KW

OLI-1H Series Hybrid Inverter

One compact unit for solar generation, 48 V battery storage and seamless backup — for homes and small businesses that need reliable power through outages and lower bills through time-of-use optimisation.

OLI-1H3L | OLI-1H3.6L | OLI-1H4L | OLI-1H4.6L | OLI-1H5L | OLI-1H6L | OLI-1H5L-2

Key figures

Rated output power	**3 – 6 kW**
Max. PV input	**up to 9.6 kW (200 %)**
MPPTs / strings	**2 / 1+1**
Battery voltage	**40 – 60 V**
Switch-over time	**< 10 ms**
Peak efficiency	**97.3 %**
Protection	**IP66, natural cooling**

- **Up to 200 % PV input** — oversize the array on two MPPTs with 18 A per string
- **Up to 200 % grid input** — fast grid charging and pass-through for heavy loads
- **48 V battery flexibility** — lithium or lead-acid, up to 135 A charge / discharge; CAN link to Olitec OLB packs
- **Parallel up to 6 units** — scale from 3 kW to 36 kW single-phase
- **Smart energy management** — 6-period time-of-use, peak shaving, app monitoring, VPP ready
- **Complete protection** — AFCI (optional), Type II SPD, insulation and leakage detection

THREE-PHASE · HIGH-VOLTAGE BATTERY · 15 – 30 KW

OLI-3H Series Hybrid Inverter

Commercial-grade three-phase hybrid with up to 60 kW of PV input, dual high-voltage battery ports and full backup output — for offices, schools, clinics, retail and light industry.

OLI-3H15H | OLI-3H20H | OLI-3H25H | OLI-3H29.9H | OLI-3H30H

Key figures

Rated output power	**15 – 30 kW**
Max. PV input	**up to 60 kW**
MPPTs / strings	**2 – 3 / 2 per MPPT**
Battery voltage	**160 – 750 V**
Switch-over time	**< 10 ms**
Peak efficiency	**97.8 %**
Protection	**IP66, fan cooling**

- **Up to 3 MPPTs, 36 A each** — 160 – 850 V MPPT window, 1000 V max. input
- **Dual battery inputs** — 50 A + 50 A on 25 kW models and above; CAN-based BMS integration
- **100 % unbalanced output** — full rated power on any single phase in backup mode
- **Parallel up to 6 units** — with AC coupling and diesel-generator integration
- **Grid-code compliant** — anti-islanding, LVRT / HVRT, residual current and insulation monitoring

OLI-1H Single-Phase Hybrid Inverter 3 – 6 kW

Low-voltage battery

Model	OLI-1H3L	OLI-1H3.6L	OLI-1H4L	OLI-1H4.6L	OLI-1H5L	OLI-1H6L	OLI-1H5L-2
DC Input (PV Side)							
Max. input power [W]	4800	5760	6400	7360	8000	9600	8000
Max. input voltage [V]	500						
Rated input voltage [V]	370						
MPPT voltage range [V]	125 – 425						
Start-up voltage [V]	125						
Max. input current per MPPT [A]	18.0 / 18.0						
Max. short-circuit current per MPPT [A]	27.0 / 27.0						
No. of MPPTs / strings per MPPT	2 / 1+1						
Battery Connection							
Battery type	Lithium-ion / Lead-acid						
Max. charge / discharge power [W]	3000 / 3000	3600 / 3600	4000 / 4000	4600 / 4600	5000 / 5000	6000 / 6000	5000 / 5000
Battery voltage range [V]	40 – 60						
Max. charge / discharge current [A]	70	90	100	110	120	135	120
AC Output (Grid Port)							
Rated output power [W]	3000	3600	4000	4600	5000	6000	5000
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Max. input apparent power from grid [VA]	6000	7200	8000	9200	10000	10000	10000
Rated grid voltage / frequency	220 / 230 V, 50 / 60 Hz						
Rated output current [A]	13.6 / 13.0	16.0 / 16.0	18.2 / 17.4	20.9 / 20.0	22.7 / 21.7	27.3 / 26.1	22.7 / 21.7
Max. output current [A]	15.0	16.0	20.0	22.7	25.0	30.0	22.7
Max. input current from grid [A]	27.3	32.7	36.4	42	43	43	43
Power factor	~1 (adjustable 0.8 leading to 0.8 lagging)						
Current THD [%]	< 3 @ rated power						
AC Output (Backup Port)							
Rated power [W]	3000	3600	4000	4600	5000	6000	5000
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Rated output voltage / frequency	220 / 230 V, 50 / 60 Hz						
Rated output current [A]	13.6 / 13.0	16.4 / 15.7	18.2 / 17.4	20.9 / 20.0	22.7 / 21.7	27.3 / 26.1	22.7 / 21.7
Max. output current [A]	15.0	18.2	20.0	22.7	25.0	30.0	22.7
Max. output peak power [W] / duration	6000 (<10 s)	7200 (<10 s)	8000 (<10 s)	9200 (<10 s)	10000 (<10 s)	10000 (<10 s)	10000 (<10 s)
Grid-tied to stand-alone switching time [ms]	< 10						
Output voltage THD [%]	< 3 @ linear load						
Efficiency							
Max. efficiency [%]	97.0	97.0	97.0	97.3	97.3	97.3	97.3
Euro efficiency [%]	96.0	96.0	96.0	96.2	96.2	96.2	96.2
Protection							
Safety protection features	LVRT, HVRT, anti-islanding, AC output over / under-voltage, AC short-circuit, AC over-current, leakage current, Type II DC / AC surge protection, PV insulation resistance detection, PV reverse polarity, PV input over-voltage, earthing fault protection, DC arc-fault detection (optional)						
General Data							
Dimensions (W × H × D) [mm]	228 × 470 × 340						
Weight [kg]	19						
Operating temperature range [°C]	–20 to +60 (derating above +45)						
Relative humidity / max. altitude	0 – 100 % (non-condensing) / 2000 m						
Ingress protection	IP66						
Cooling	Natural convection						
User interface	LCD, LED, mobile app						
Communication with BMS	CAN						
Typical noise emission [dB]	< 35						
Standards & Warranty							
Compliance	IEC 62109-1 & -2, IEC 61000-6-1, IEC 61000-6-3, IEC/EN 62920, IEC 60068, IEC 61683, IEC 63027, IEC 60730, EN 303645, IEC 61727, IEC 62116, EN 50549-1 / EN 50549-10, IS 16221 / IS 16169 / IS 17980						
Standard warranty [years]	5 (extended warranty available)						

OLI-1H5L-2: 5 kW variant without 10 % AC-side overload capacity. Rated output currents shown at 220 V / 230 V. Warranty per Olitec Solar warranty policy. Specifications subject to change without notice.

OLI-3H Three-Phase Hybrid Inverter 15 – 30 kW

High-voltage battery

Model	OLI-3H15H	OLI-3H20H	OLI-3H25H	OLI-3H29.9H	OLI-3H30H
DC Input (PV Side)					
Max. input power [W]	30000	40000	50000	59800	60000
Max. input voltage [V]			1000		
Rated input voltage [V]			650		
MPPT voltage range [V]			160 – 850		
Start-up voltage [V]			180		
Max. input current per MPPT [A]	36 / 36	36 / 36	36 / 36 / 36	36 / 36 / 36	36 / 36 / 36
Max. short-circuit current per MPPT [A]	45 / 45	45 / 45	45 / 45 / 45	45 / 45 / 45	45 / 45 / 45
No. of MPPTs / strings per MPPT	2 / 2+2	2 / 2+2	3 / 2+2+2	3 / 2+2+2	3 / 2+2+2
Battery Connection					
Battery type			Lithium-ion (high-voltage)		
Max. charge / discharge power [W]	16500	22000	27500	30000	33000
Battery voltage range [V]			160 – 750		
Max. charge / discharge current [A]	50	50	50 + 50	50 + 50	50 + 50
Number of battery inputs	1	1	2	2	2
AC Output (Grid Port)					
Rated output power [W]	15000	20000	25000	29900	30000
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Max. input apparent power from grid [VA]	30000	40000	50000	60000	60000
Rated grid voltage / frequency			3L / N / PE, 220 / 380, 230 / 400 V, 50 / 60 Hz		
Rated output current [A]	22.8 / 21.8	30.4 / 29	37.9 / 36.3	45.4 / 43.4	45.4 / 43.5
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
Max. input current from grid [A]	45.5 / 43.3	60.6 / 57.7	76.0 / 72.2	90 / 86.6	90 / 86.6
Power factor			~1 (adjustable 0.8 leading to 0.8 lagging)		
Current THD [%]			< 3 @ rated power		
AC Output (Backup Port)					
Rated power [W]	15000	20000	25000	29900	30000
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Rated output voltage / frequency			3L / N / PE, 220 / 380, 230 / 400 V, 50 / 60 Hz		
Rated output current [A]	22.8 / 21.8	30.4 / 29	37.9 / 36.3	45.4 / 43.4	45.4 / 43.5
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
Max. output peak power [W] / duration	30000 (<10 s)	40000 (<10 s)	40000 (<10 s)	48000 (<10 s)	48000 (<10 s)
Grid-tied to stand-alone switching time [ms]			< 10		
Output voltage THD [%]			< 3 @ linear load		
Efficiency					
Max. efficiency [%]			97.8		
Euro efficiency [%]	96.7	97.2	97.2	97.2	97.2
Protection					
Safety protection features	Anti-islanding, DC reverse polarity, insulation monitoring, residual current monitoring, Type II DC / AC surge protection, DC arc-fault detection (optional), three-phase unbalanced output				
General Data					
Dimensions (W × H × D) [mm]			550 × 800 × 230		
Weight [kg]	46.5	46.5	50	50	50
Operating temperature range [°C]			–25 to +60		
Relative humidity / max. altitude			0 – 100 % / 3000 m		
Ingress protection			IP66		
Cooling			Intelligent fan cooling		
User interface			LCD, LED, mobile app		
Communication with BMS			CAN		
Typical noise emission [dB]	< 55	< 55	< 60	< 60	< 60
Standards & Warranty					
Compliance	IEC/EN 62109-1 & -2, IEC/EN 61000-6-1 / -2 / -3 / -4, IEC/EN 62920, IEC 60068-2-1 / -2 / -14 / -27 / -30 / -64, IEC 61683, EN 50530, EN 50549-1 & -10, IEC 61727, IEC 62116, NRS 097-2-1				
Standard warranty [years]			5 (extended warranty available)		

Rated output currents shown at 220 V / 230 V phase voltage. Warranty per Olitec Solar warranty policy. Specifications subject to change without notice.